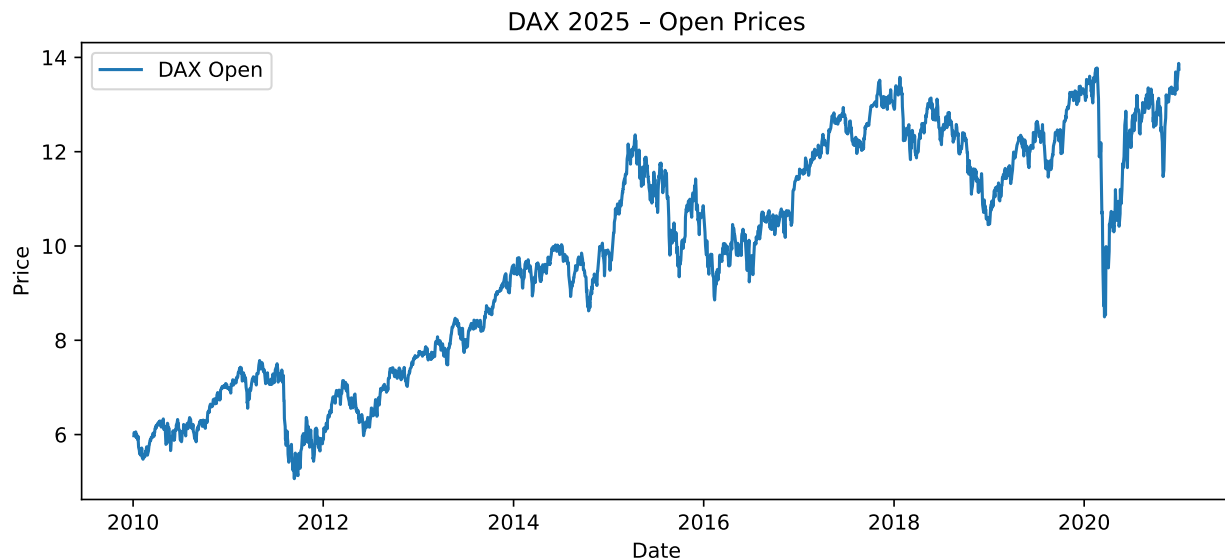


STATS 531 - Midterm Project

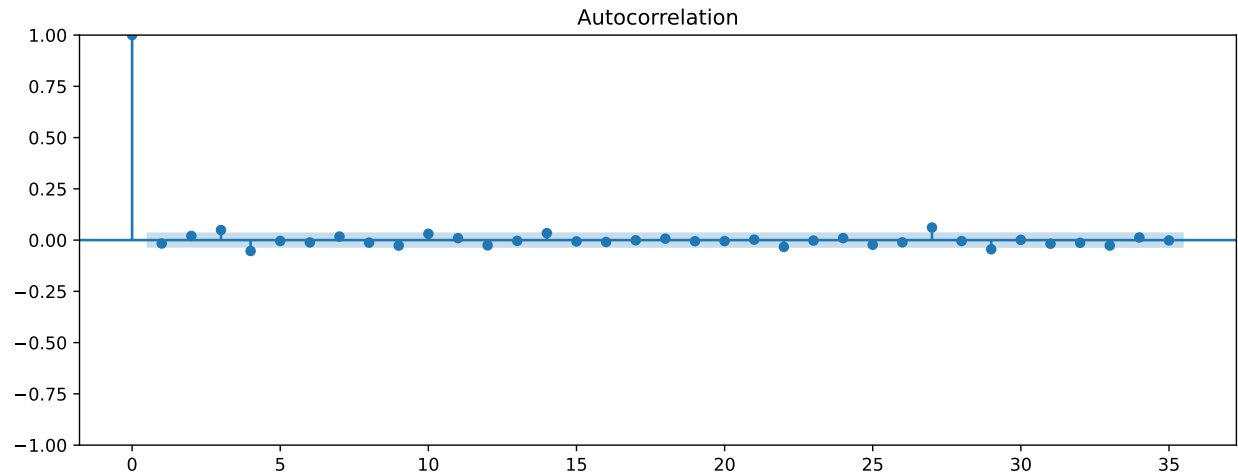
Steven Anthony Leone

Ethan Naegele

We look into the DAX Stock Index in Germany in this project. We look at the daily stock price for the DAX index for every day in 2025 from a time series analysis perspective, and we see how accurate we can make a model that predicts future values based off of its lags.

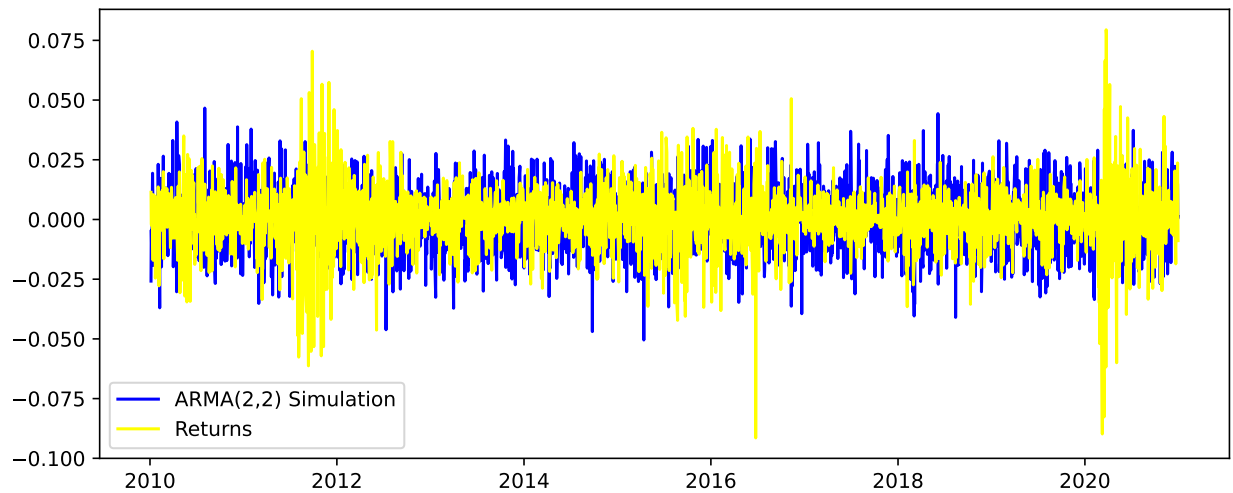


The data above shows roughly nonstationary data. We look at the autocorrelation function below to see how much lags depend on each other. In general, what we can tell is that the first point has a high correlation at the start - this is when lag is 0, so, it is, naturally high. The values that follow seem i.i.d..



ARMA

Let's train a model to accurately reflect the data. ARMA models do not work well with nonstationary data; we counter this with an ARIMA model that handles differencing.

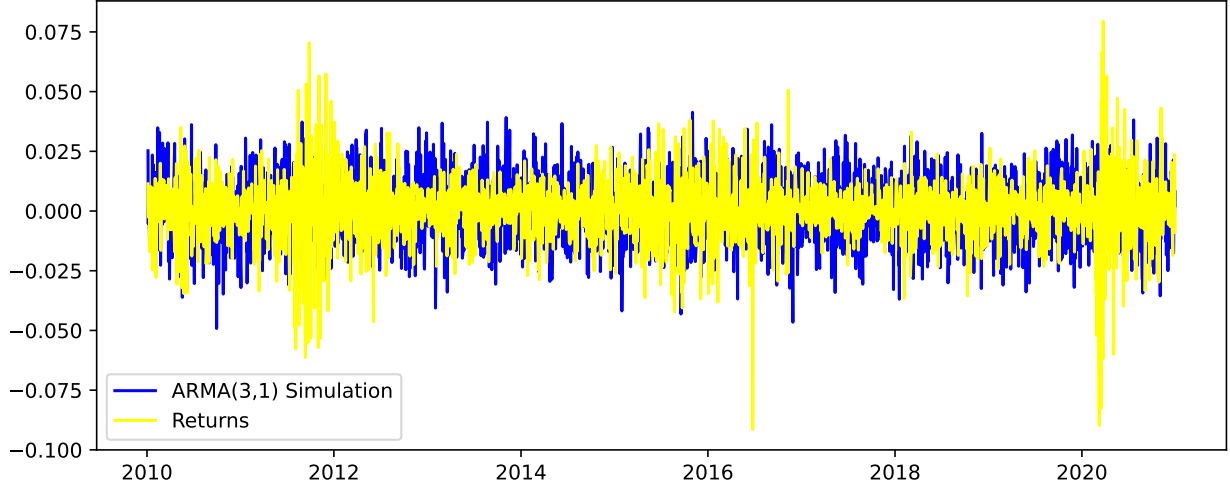


We look at the AIC values of the ARMA models below. The smallest AIC value is -3961.6, for ARMA (3, 1).

We graph an ARMA(3, 1) model below. It is not much different from ARMA(2, 2).

	MA(1)	MA(2)	MA(3)	MA(4)
AR(1)	-3965.5	-3965.7	-3972.9	-3973.5
AR(2)	-3962.0	-3966.6	-3972.8	-3971.7
AR(3)	-3961.6	-3970.7	-3974.7	-3972.2
AR(4)	-3972.8	-3974.8	-3972.4	-3968.0

Table 1: ARMA Model Comparison



SARMA

The ARMA model can be written with a backshift operator, B , to represent the previous lag in a sequence. We can multiply a sequence by B in order to represent a lag farther back.

$$By_n = y_{n-1}$$

$$B^2y_n = y_{n-2}$$

$$B^{12}y_n = y_{n-12}$$

When we have an ARMA model with white noise ϵ_t , we can represent the model simply with backshift operators and $\phi(B) = (1 - \phi_1 * B - \phi_2 * B^2 - \dots \phi_p * B^p)$.

$$y_n = \phi_1 * y_{n-1} + \phi_2 * y_{n-2} + \dots \phi_p * y_{n-p} + \epsilon_n$$

$$y_n - (\phi_1 * y_{n-1} + \phi_2 * y_{n-2} + \dots \phi_p * y_{n-p}) = \epsilon_n$$

$$y_n - (\phi_1 * B * y_n + \phi_2 * B^2 * y_n + \dots \phi_p * B^p * y_n) = \epsilon_n$$

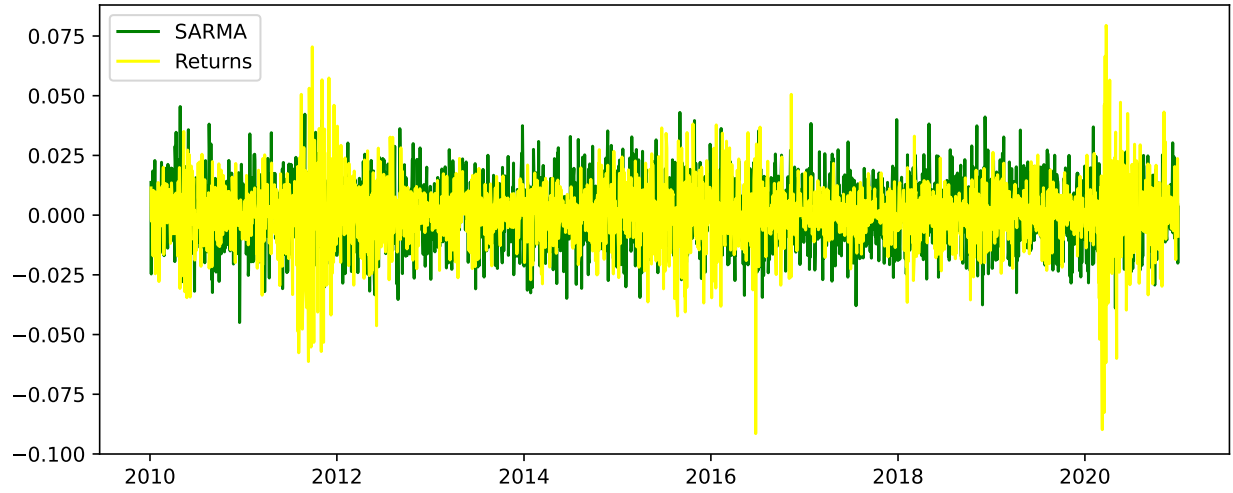
$$(1 - \phi_1 * B - \phi_2 * B^2 - \dots \phi_p * B^p)y_n = \epsilon_n$$

$$\phi(B)y_n = \epsilon_n$$

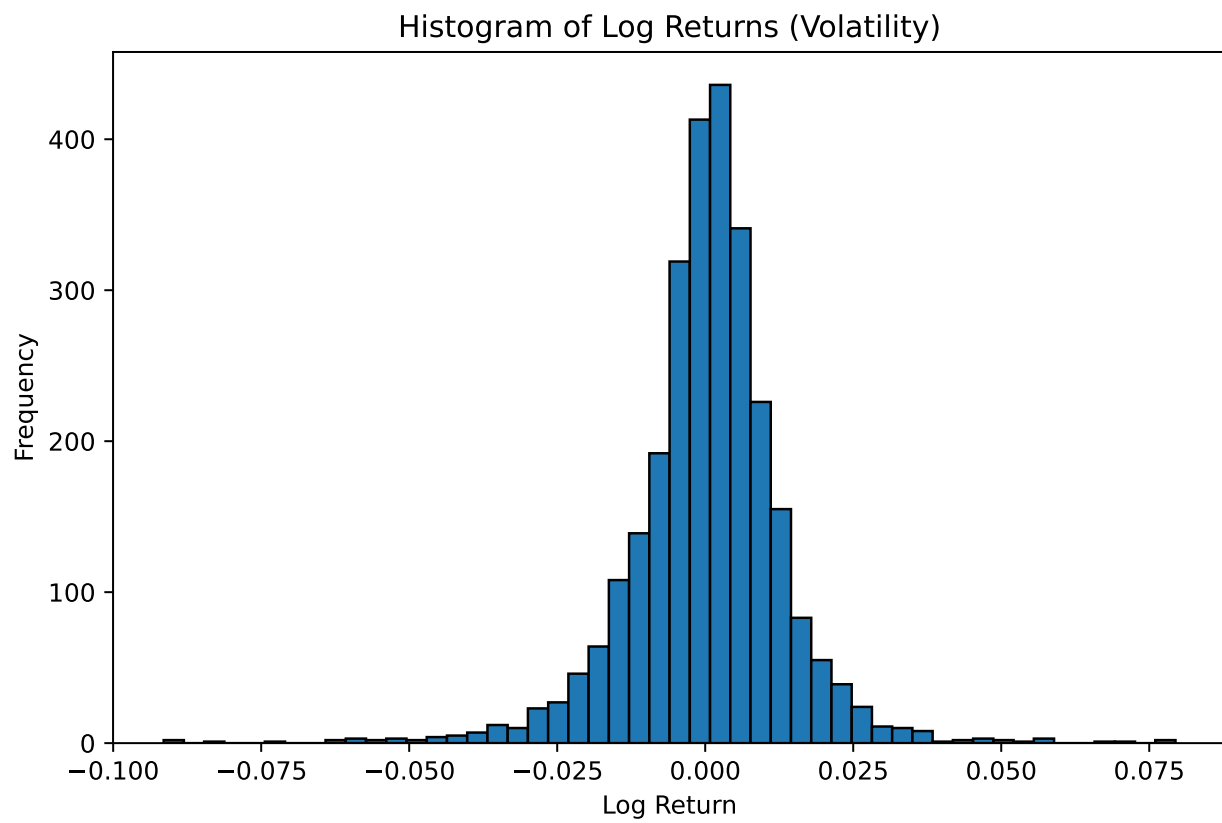
In a seasonal model, we may only represent seasonal terms, ex, every 12. We can easily do this with the backshift operator with four functions, representing it as

$$\Phi(B^{12}) * \phi(B)y_t = \theta(B) * \theta(B^{12}) * \epsilon_t$$

This example will give us regular lags, as well as lags for every 12 (seasonal lags). As stock changes prices rapidly, this is relevant to a DAX index. We test it out below.



Volatility histogram



TODO ...

References